

Deep Learning and Its Application

Week 3: Practicals

1. Given

```
from sklearn.neural_network import MLPClassifier
from sklearn.datasets import make_classification
from sklearn.model_selection import train_test_split
X, y = make_classification(n_samples=100, random_state=1)
```

Develop a classification model using Multi Perceptron class **MLPClassifier** in Scikit_Learn and display the probabilities $P(y|x)$, prediction results, and scores, loss function, activation function

2. Given

```
from sklearn.neural_network import MLPClassifier
from sklearn.datasets import make_regression
from sklearn.model_selection import train_test_split
X, y = make_regression(n_samples=200, random_state=1)
```

Develop a classification model using Multi Perceptron class **MLPRegression** in Scikit_Learn and display prediction results, and scores, loss function, activation function

3. Given

```
from numpy import argmax
from pandas import read_csv
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from tensorflow.keras import Sequential
from tensorflow.keras.layers import Dense
import numpy as np

# load the dataset
path = 'https://raw.githubusercontent.com/jbrownlee/Datasets/master/iris.csv'
df = read_csv(path, header=None)
# split into input and output columns
X, y = df.values[:, :-1], df.values[:, -1]
# ensure all data are floating point values
X = X.astype('float32')
# encode strings to integer
y = LabelEncoder().fit_transform(y)
# split into train and test datasets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.33)
print(X_train.shape, X_test.shape, y_train.shape, y_test.shape)
# determine the number of input features
n_features = X_train.shape[1]

.
.
.

row2 = np.array([[5.1, 3.5, 1.4, 0.2]])

yhat = model.predict([row2])
print('Predicted: %s (class=%d)' % (yhat, argmax(yhat)))
```

Use **Sequential** model to develop a multiclass classification model using Iris dataset, display

- a) Training accuracy
- b) Prediction results

4. Given

```
from numpy import sqrt
from pandas import read_csv
from sklearn.model_selection import train_test_split
from tensorflow.keras import Sequential
from tensorflow.keras.layers import Dense
import numpy as np

# load the dataset
path =
'https://raw.githubusercontent.com/jbrownlee/Datasets/master/housing.csv'
df = read_csv(path, header=None)
# split into input and output columns
X, y = df.values[:, :-1], df.values[:, -1]
# split into train and test datasets
X_train, X_test, y_train, y_test =
.
.
.

row2 =
np.array([[0.00632, 18.00, 2.310, 0, 0.5380, 6.5750, 65.20, 4.0900, 1, 296.0, 15.30, 396.9
0, 4.98]])
yhat = model.predict([row2])
print('Predicted: %.3f' % yhat)
```

Use **Sequential** model to develop a regression model using Boston Housing dataset, display

- c) Squared sum error
- d) Prediction results